

KESTRELATTICE

Governed Agent Mesh | Micro-Product

AI Model Rotation Policy One-Pager

One page that decides when your agents move to a new model version -- and who signs off first. A complete, zero-placeholder draft for teams governing AI agents in production: policy gates, audit trails, cost controls, permissioning, and incident response, written so you can adopt it as-is.

A complete, ready-to-use governance document. Adopt it as-is, customize the bracketed fields to your environment, and put it to work today.

Instant delivery | Yours to keep, forever.

AI MODEL ROTATION POLICY ONE-PAGER

Version: [VERSION NUMBER] **Effective Date:** [EFFECTIVE DATE] **Authorized By:** [TEAM NAME] Leadership **Policy Owner:** [POLICY OWNER NAME] **Approved By:** [APPROVAL AUTHORITY ROLE]

1. PURPOSE

This policy defines the governance framework for rotating AI models in production to ensure **security, cost efficiency, performance stability, and compliance** with [ORGANIZATION NAME]'s AI governance standards. It applies to all AI agents deployed under the purview of [TEAM NAME], including but not limited to: - [List of AI systems/agents, e.g., "Customer Support Chatbots," "Fraud Detection Models," "Recommendation Engines"] - Third-party integrated models accessed via [API NAME] or [PLATFORM NAME] - Custom fine-tuned models hosted on [INFRASTRUCTURE NAME]

Non-compliance with this policy may result in **deployment freeze, cost recovery, or disciplinary action** per [EMPLOYEE HANDBOOK SECTION].

2. SCOPE

2.1 Applicable Models

All AI models in **production, staging, or pre-production** environments must adhere to this policy, including: - **Base models** (e.g., [MODEL NAME] from [PROVIDER NAME]) - **Fine-tuned variants** (e.g., [CUSTOM MODEL NAME] trained on [DATASET NAME]) - **Hybrid models** (e.g., ensemble systems combining [MODEL TYPE 1] and [MODEL TYPE 2])

Exclusions: - [List any exceptions, e.g., "Legacy models in read-only mode pre-[DATE]"] - Models governed by **separate compliance frameworks** (e.g., [REGULATORY NAME] for healthcare)

2.2 Responsibilities

Role	Responsibilities
[TEAM NAME] Lead	Approves rotation schedules, escalates risks, and ensures cost controls.
Model Owner	Submits rotation requests, documents changes, and verifies post-rotation performance.
Security Lead	Validates model integrity, patch compliance, and access controls.
Cost Accountant	Monitors API usage, enforces budget caps, and flags anomalies.
Incident Response	Investigates failures, reverts models, and documents lessons learned.

3. ROTATION TRIGGERS & SCHEDULE

3.1 Mandatory Rotation Events

Rotate models **immediately** upon: 1. **Security Vulnerabilities:** - Public disclosure of a critical flaw in the base model (e.g., [CVE NAME]). - Detection of **data poisoning** or **adversarial attacks** (per [THREAT MODELING DOC]). - **Third-party provider update** requiring compliance (e.g., [PROVIDER NAME]'s [VERSION] patch).

1. **Performance Degradation:**
2. **Accuracy drop** > [PERCENTAGE] (measured via [METRIC NAME], e.g., "BLEU score") over [TIME WINDOW].
3. **Latency spike** > [PERCENTAGE] (e.g., "50% increase in P99 response time").

Cost per query > [\$AMOUNT] (triggering [BUDGET ALERT THRESHOLD]).

Compliance Changes:

6. New regulations (e.g., [REGULATION NAME]) requiring model updates.

Bias audits revealing non-compliance with [BIAS METRIC NAME] thresholds.

Provider Deprecations:

9. **End-of-life (EOL) announcement** from [PROVIDER NAME] for [MODEL NAME].
10. **API deprecation** without a supported alternative.

3.2 Scheduled Rotations

Frequency	Trigger	Approval Required	Audit Trail Required
Quarterly	[PROVIDER NAME]'s recommended updates for [MODEL NAME].	[TEAM NAME] Lead	Yes (via [AUDIT TOOL NAME])
Annually	Major version bumps (e.g., [MODEL NAME] v[NEW_VERSION]).	[APPROVAL AUTHORITY ROLE]	Yes (with cost-benefit analysis)
On-Demand	Model Owner request (e.g., for [REASON, e.g., "improved few-shot support"]).	[REQUESTER'S MANAGER]	Yes (with justification)

Note: Scheduled rotations must be **backward-compatible** unless explicitly approved for a **canary deployment**.

4. ROTATION PROCESS

4.1 Pre-Rotation Checklist

Before initiating rotation, the **Model Owner** must: 1. **Verify Compatibility:** - Confirm new model supports [REQUIRED FEATURES, e.g., "multilingual input," "input length limits"]. - Test against [REGRESSION TEST SUITE NAME] with **>95% pass rate**.

1. **Cost Estimation:**
2. Compare **TCO (Total Cost of Ownership)** of old vs. new model using [COST CALCULATOR NAME].

Ensure new model does not exceed [BUDGET CAP] for [ROTATION CYCLE].

Access & Permissions:

5. Update **IAM roles** for [MODEL NAME] via [ACCESS MANAGEMENT TOOL].

Revoke access to deprecated models per [DATA RETENTION POLICY].

Incident Response Plan:

8. Define **rollback criteria** (e.g., "if error rate > [PERCENTAGE] in [TIME WINDOW]").

Assign **on-call responder** from [INCIDENT TEAM NAME].

Documentation Update:

11. Update [MODEL DOCUMENTATION LINK] with:
 - New **input/output schemas**.
 - **Known limitations** (e.g., "fails on inputs > [CHARACTER LIMIT]").
 - **Deprecation timeline** for old model.

4.2 Rotation Execution

1. Canary Deployment (Mandatory for Production):

2. Route [PERCENTAGE] of traffic (e.g., "5%") to new model via [ROUTING TOOL NAME].

Monitor for [TIME WINDOW] (e.g., "24 hours") using [MONITORING DASHBOARD NAME].

Full Rollout:

5. If canary passes, proceed to **full deployment** via [DEPLOYMENT TOOL NAME].

Disable old model after confirmation of new model stability.

Post-Rotation Validation:

8. Run [AUDIT SCRIPT NAME] to verify:
 - No **data leakage** from deprecated model.
 - **Compliance** with [PRIVACY POLICY NAME].
 9. Log results in [AUDIT DATABASE NAME].
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5. COST & BUDGET CONTROLS

5.1 Budget Enforcement

- **Daily Spending Limit:** [AMOUNT] per model per [TIME PERIOD].
- **Alert Thresholds:**
- **Warning:** Usage exceeds [PERCENTAGE] of limit (e.g., "80%").

Emergency: Usage exceeds [PERCENTAGE] (e.g., "120%") -> **automatic pause** until approval from [COST ACCOUNTANT ROLE].

Cost Allocation:

- Tag all API calls with [PROJECT_NAME]-[MODEL_NAME] for chargeback via [FINANCE SYSTEM NAME].

5.2 Provider Contracts

- Ensure contracts include:
- **Cost guarantees** for model rotations (e.g., "no hidden fees for [MODEL NAME] v[NEW_VERSION]").
- **SLA credits** for downtime during mandatory updates.

6. SECURITY & COMPLIANCE

6.1 Model Integrity

- **Cryptographic Verification:**
- Use [TOOL NAME] to validate model weights against [PROVIDER NAME]'s signed hashes.
- **Input Sanitization:**
- Enforce [INPUT VALIDATION RULES] (e.g., "reject prompts > [TOKEN LIMIT]").

6.2 Audit Trail Requirements

Every rotation must generate: 1. **Change Log:** Timestamped record in [AUDIT SYSTEM NAME] with: - **Old vs. new model versions.** - **Approving authority.** - **Cost impact** (old vs. new). 2. **Performance Metrics:** Pre/post-rotation [METRIC NAME] scores. 3. **Access Logs:** Who had permissions to trigger rotation (via [LOG SYSTEM NAME]).

Retention: Audit records must be retained for [YEARS] per [DATA RETENTION POLICY].

7. INCIDENT RESPONSE

7.1 Failure Scenarios & Actions

Scenario	Detection	Escalation Path	Mitigation
Model Crash	[MONITORING TOOL NAME] alerts	[INCIDENT NAME] (SLA: [TIME]) TEAM	Revert to last stable version via [COMMAND].
Bias/Toxicity Violation	[AUDIT TOOL NAME] flag	[ETHICS BOARD NAME]	Deploy mitigation ruleset; log incident.
Cost Spiral	[BUDGET ALERT] triggers	[COST ACCOUNTANT ROLE]	Pause API access; review usage patterns.
Data Leak	[SECURITY TOOL NAME] detects	[SECURITY LEAD] (immediate)	Isolate model; investigate via [INVESTIGATION PROCESS].

7.2 Post-Incident Review

- **Root Cause Analysis (RCA):** Completed within [DAYS] via [RCA FORM NAME].
- **Lessons Learned:** Documented in [LESSONS LEARNED DATABASE] and shared with [TEAM NAME].
- **Retraining:** If applicable, update [MODEL TRAINING PIPELINE] per [IMPROVEMENT PLAN].

8. DOCUMENTATION & ADOPTION NOTES

8.1 Required Documentation

All buyer-fillable fields in this policy must be: 1. **Fully populated** using the [BRACKETED CAPS] convention. 2. **Documented in the adoption notes** below with: - **Justification** (e.g., "Quarterly rotations align with [PROVIDER NAME]'s update cycle"). - **Owner** (e.g., "[TEAM NAME] Lead approves all scheduled rotations"). - **Tools/Processes** (e.g., "Cost monitored via [FINANCE SYSTEM NAME]").

**8.2 Adoption

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The governed agent mesh for production AI.

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